Password Entropy Calculator
Calculator: password-entropy-calculator | Engine version: 1
Raw engine values: full numeric precision; no display rounding or sampled rows.
Units appear only where supplied by the calculation or bound display metadata.
Embedded Unicode glyphs are readable.
Unsupported characters and controls use explicit Unicode escapes.
Tagged PDF; PDF/UA conformance: UNVERIFIED.
Text follows document order; charts include complete numeric text alternatives.

INPUTS
Length
input["length"] = 16
Character pool size
input["characterPoolSize"] = 94

RESULTS
Theoretical entropy [bits]
output["value"] = 104.8734216268422
output["steps"][0] = "Password Entropy = 104.8734216268422"

METHODOLOGY
Deterministic calculation using the reviewed Technology & Computing engine.\u000aTechnical calculation revision: technology.password-entropy@1\u000a\u000aTheoretical entropy for uniformly random independent characters drawn from the supplied pool; not an estimate for human-chosen passwords.\u000aEntropy bits = length × log2(character pool size)

DECLARED FORMULAS
password-entropy-calculator: Entropy bits = length × log2(character pool size)
Theoretical entropy for uniformly random independent characters drawn from the supplied pool; not an estimate for human-chosen passwords.

SOURCES
NIST SP 800-63B — Passwords
https://pages.nist.gov/800-63-4/sp800-63b/passwords/
NIST cautions that entropy for user-chosen passwords is difficult to estimate; this calculator is only the theoretical uniform-random model.